

Q1.

- (a) A discrete random variable X has a probability function defined by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, 3, 4, \dots$$

- (i) State the name of the distribution of X .

(1)

- (ii) Write down, in terms of λ , expressions for $E(3X - 1)$ and $\text{Var}(3X - 1)$.

(2)

- (iii) Write down an expression for $P(X = x + 1)$, and hence show that

$$P(X = x + 1) = \frac{\lambda}{x + 1} P(X = x)$$

(3)

- (b) The number of cars and the number of coaches passing a certain road junction may be modelled by independent Poisson distributions.

- (i) On a winter morning, an average of 500 cars per hour and an average of 10 coaches per hour pass this junction.

Determine the probability that a total of at least 10 such vehicles pass this junction during a particular 1-minute interval on a winter morning.

(3)

- (ii) On a summer morning, an average of 836 cars per hour and an average of 22 coaches per hour pass this junction.

Calculate the probability that a total of at most 3 such vehicles pass this junction during a particular 1-minute interval on a summer morning. Give your answer to two significant figures.

(3)

(Total 12 marks)

Q2.

The random variable Y has a Poisson distribution with $E(Y) = 2.5$.

Given that $Z = 4Y + 30$:

- (a) show that $\text{Var}(Z) = E(Z)$;

(3)

- (b) give a reason why the distribution of Z is **not** Poisson.

(1)

(Total 4 marks)

Q3.

John works from home. The number of business letters, X , that he receives on a weekday may be modelled by a Poisson distribution with mean 5.0.

The number of private letters, Y , that he receives on a weekday may be modelled by a Poisson distribution with mean 1.5.

- (a) Find, for a given weekday:
- (i) $P(X < 4)$; (2)
 - (ii) $P(Y = 4)$. (2)
- (b) (i) Assuming that X and Y are independent random variables, determine the probability that, on a given weekday, John receives a **total** of more than 5 business and private letters. (3)
- (ii) Hence calculate the probability that John receives a **total** of more than 5 business and private letters on at least 7 out of 8 given weekdays. (3)
- (c) The numbers of letters received by John's neighbour, Brenda, on 10 consecutive weekdays are
- 15 8 14 7 6 8 2 8 9 3
- (i) Calculate the mean and the variance of these data. (2)
 - (ii) State, giving a reason based on your answers to part (c)(i), whether or not a Poisson distribution might provide a suitable model for the number of letters received by Brenda on a weekday. (2)

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(Total 14 marks)

Q4.

A new information technology centre is advertising places on its one-week residential computer courses.

- (a) The number of places, X , booked each week on the publishing course may be modelled by a Poisson distribution with a mean of 9.0.
- (i) State the standard deviation of X . (1)
 - (ii) Calculate $P(6 < X < 12)$. (3)
- (b) The number of places booked each week on the web design course may be modelled by a Poisson distribution with a mean of 2.5.
- (i) Write down the distribution for T , the **total** number of places booked each

week on the publishing and web design courses.

(1)

- (ii) Hence calculate the probability that, during a given week, a total of fewer than 2 places are booked.

(3)

- (c) The number of places booked on the database course during each of a random sample of 10 weeks is as follows:

14 15 8 16 18 4 10 12 15 8

By calculating appropriate numerical measures, state, with a reason, whether or not the Poisson distribution $Po(12.0)$ could provide a suitable model for the number of places booked each week on the database course.

(3)

(Total 11 marks)

Q5.

The number of computers, A , bought during one day from the Amplebuy computer store can be modelled by a Poisson distribution with a mean of 3.5.

The number of computers, B , bought during one day from the Bestbuy computer store can be modelled by a Poisson distribution with a mean of 5.0.

- (a) (i) Calculate $P(A = 4)$.

(2)

- (ii) Determine $P(B \leq 6)$.

(1)

- (iii) Find the probability that a total of fewer than 10 computers is bought from these two stores on one particular day.

(3)

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- (b) Calculate the probability that a total of fewer than 10 computers is bought from these two stores on at least 4 out of 5 consecutive days.

(3)

- (c) The numbers of computers bought from the Choicebuy computer store over a 10-day period are recorded as

8 12 6 6 9 15 10 8 6 12

- (i) Calculate the mean and variance of these data.

(2)

- (ii) State, giving a reason based on your results in part (c) (i), whether or not a Poisson distribution provides a suitable model for these data.

(2)

(Total 13 marks)

Q6.

The administration department at Hitech College uses two photocopiers, A and B, which operate independently.

The number of breakdowns each week, X , for photocopier A can be modelled by a Poisson distribution with a mean of 0.5.

The number of breakdowns each week, Y , for photocopier B can be modelled by a Poisson distribution with a mean of 2.5.

(a) Calculate:

(i) $P(X \leq 1)$;

(2)

(ii) $P(Y = 2)$.

(2)

(b) (i) In a given week, show that the probability of a total of exactly 2 photocopier breakdowns is 0.224, correct to three decimal places.

(3)

(ii) Hence find the probability that there will be exactly 2 photocopier breakdowns in each of four consecutive weeks.

(2)

(c) The total number of photocopier breakdowns, T , in a 4-week period can be modelled by a Poisson distribution with mean μ .

(i) Find the value of μ .

(1)

(ii) Hence calculate the probability that, in a given 4-week period, there will be a total of at least 18 photocopier breakdowns.

(2)

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(Total 12 marks)

Q7.

At a petrol station the number of 'drive-outs' (motorists who obtain petrol and then drive away without paying) may be modelled by a Poisson distribution with mean 0.8 per day.

(a) Find the probability that, on a particular day, the number of 'drive-outs' is:

(i) 2 or fewer;

(ii) more than 1 but fewer than 3.

(4)

(b) (i) Evaluate the mean and the standard deviation of the number of 'drive-outs' during a 5-day period.

(3)

(ii) Find the probability that there are fewer than 3 'drive-outs' during a 5-day period.

(1)

Q8.

In the Parkside Sunday Football League, the number of goals scored in each match may be modelled by Poisson distributions.

- (a) When the weather is dry, the mean number of goals scored is 1.6 per 60 minutes of play.

Find the probability that on a dry Sunday:

- (i) exactly 3 goals are scored during the first 60 minutes of a particular match; (2)

- (ii) a match lasting **90 minutes** ends with the score 0–0; (2)

- (iii) when 5 matches, each lasting 90 minutes, are played, a total of 20 or more goals is scored. (3)

- (b) When the weather is wet, the mean number of goals scored is 1.2 per 60 minutes of play.

For a match lasting 90 minutes played on a wet Sunday:

- (i) find the standard deviation of the number of goals scored; (2)

- (ii) explain why there is insufficient data to calculate the probability of the final score being 3–0. (1)

- (c) A statistician plans to examine the distribution of the number of goals scored in all matches played during the 2004/5 season. Explain why a Poisson distribution is unlikely to provide an adequate model for this distribution.

(1)

(Total 11 marks)

Q9.

The number of smoke alarms sold by a store may be modelled by a Poisson distribution with mean 7.5 per day.

- (a) Find the probability that during a particular day the store sells:

- (i) 6 or fewer smoke alarms;

- (ii) exactly 8 smoke alarms.

(4)

- (b) Evaluate the standard deviation of the number of smoke alarms sold by the store per day.

(2)

(Total 6 marks)

Q10.

- (a) Suggest a distribution which might be appropriate for modelling the number of new bicycles sold per day by a bicycle shop.

(1)

- (b) The shop sells the following numbers of new bicycles on successive days.

4 9 1 6 3 4 6 4 5 1 0 5 0

- (i) Calculate the mean and the variance of these data.

(2)

- (ii) State, giving a reason, whether your calculations in part (b)(i) support the model you have suggested in part (a).

(2)

(Total 5 marks)



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